Theory of Computation-PROBLEM SET 5

June 2, 2026

Due June 17th. Upload to learn.tsinghua.edu.cn

Problem 1

Show that the complete bipartite graph $K_{3,3}$ has no Pfaffian orientation.

Problem 2

(*"Were-you-last" game*) Can you improve the update time of the decremental counter introduced in class to $4 \log \log m + O(1)$? (Suppose $m = 2^i$ for some integer i.) Describe your method.

Problem 3

Prove that the APSP conjecture implies that the following problem has no $O(n^{3-\varepsilon})$ time algorithm for any constant $\varepsilon > 0$: (Here the *absolute values* of all edge weights and matrix elements are bounded by M, a polynomial in n.)

- You are given a two dimensional array A of size $n \times n$, where each element is an integer between $[-M, M]$.
- The goal is to find the continuous sub-array with maximum total sum, i.e. to find $1 \leq i \leq i' \leq n$ and $1 \leq j \leq j' \leq n$ such that

$$\sum_{a=i}^{i'} \sum_{b=j}^{j'} A_{a,b}$$

  is maximized.

Problem 4 (Optional)

We have seen that if reversible programs P_l, P_r can cleanly compute values v_l, v_r into registers T_l, T_r, respectively, for values and registers in field $\mathcal{R}$, with $m = |\mathcal{R}| - 1 > 1$, then for a polynomial $f(y, z)$ over $\mathcal{R}$ of degree $< m$, $f(v_l, v_r)$ can be cleanly computed into register T_u by $4m$ recursive calls to $P_l, P_r, P_l^{-1}, P_r^{-1}$. (It is a simplified version of class material, as f only takes two inputs from $\mathcal{R}$, but it is not multilinear.) Show that we just need $2m$ recursive calls to cleanly compute $f(v_l, v_r)$ into T_u.